Samet Demir

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- sametdemir.xyz • github.com/simitii • Google Scholar

Research Interests

Learning theory, feature learning, in-context learning, high-dimensional statistics, random matrix theory, universality, robust learning

Education

- **Ph.D., Computer Science and Engineering** Aug 2020 – Aug 2026
  Koc University, Turkey (Advisor: Prof. Zafer Dogan)
- **B.S., Computer Engineering** Aug 2015 – Aug 2020
  Bogazici University, Turkey

Publications

Peer-reviewed

- **Samet Demir** and Zafer Dogan. *Decomposing SGD Dynamics in Neural Networks: Teacher-Induced Spikes and Variance Inflation.* In *Advances in Neural Information Processing Systems* ***(NeurIPS)***, 2026.
- **Samet Demir** and Zafer Dogan. *Input-Label Correlation Governs a Linear-to-Nonlinear Transition in Random Features under Spiked Covariance.* *SIAM Journal on Mathematics of Data Science* ***(SIMODS)***, Vol. 8, No. 4, pp. 1322–1349, 2026.
- **Samet Demir** and Zafer Dogan. *Optimal Attention Temperature Improves the Robustness of In-Context Learning under Distribution Shift in High Dimensions.* In *Proceedings of the International Conference on Machine Learning* ***(ICML)***, 2026.
- **Samet Demir** and Zafer Dogan. *How Data Mixing Shapes In-Context Learning: Asymptotic Equivalence for Transformers with MLPs.* In *Advances in Neural Information Processing Systems* ***(NeurIPS)***, 2025.
- **Samet Demir** and Zafer Dogan. *Asymptotic Analysis of Two-Layer Neural Networks after One Gradient Step under Gaussian Mixtures with Structure.* In *Proceedings of International Conference on Learning Representations* ***(ICLR)***, 2025.
- E. Ilke Genc, **Samet Demir**, Zafer Dogan. *Learnability and Competition in High-Dimensional Multi-Component ICA.* In *ICML 2026 Workshop on High-dimensional Learning Dynamics (HiLD)*, 2026. arXiv:2605.08552.
- Onat Ure, **Samet Demir**, Zafer Dogan. *Learning Beyond the Gaussian Data: Learning Dynamics of Neural Networks on an Expressive and Cumulant-Controllable Data Model.* In *Proceedings of the IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2026.

- M. Oguzhan Gultekin, **Samet Demir**, Zafer Dogan. *Learning Rate Should Scale Inversely with High-Order Data Moments in High-Dimensional Online Independent Component Analysis.* In *Proceedings of IEEE International Workshop on Machine Learning for Signal Processing (MLSP)*, 2025.
- Yigit E. Yildirim, **Samet Demir**, Zafer Dogan. *Benefits of Online Tilted Empirical Risk Minimization: A Case Study of Outlier Detection and Robust Regression.* In *Proceedings of IEEE International Workshop on Machine Learning for Signal Processing (MLSP)*, 2025.
- **Samet Demir** and Zafer Dogan. *Asymptotic Study of In-context Learning with Random Transformers through Equivalent Models.* In *Proceedings of IEEE International Workshop on Machine Learning for Signal Processing (MLSP)*, 2025.
- **Samet Demir** and Zafer Dogan. *Optimal Nonlinearities Improve Generalization Performance of Random Features.* In *Proceedings of the Asian Conference on Machine Learning (ACML)*, 2023.
- **Samet Demir** and Zafer Dogan. *Studying Generalization Performance of Random Feature Model through Equivalent Models.* In *Proceedings of the IEEE Signal Processing and Communications Applications Conference (SIU)*, 2024.
- **Samet Demir**, Hasan F. Eniser, Alper Sen. *DeepSmartFuzzer: Reward Guided Test Generation For Deep Learning.* In *Proceedings of the IJCAI 2020 Workshop on Artificial Intelligence Safety (AISafety 2020)*, 2020.

Preprints

- **Samet Demir** and Zafer Dogan. *Implicitly Normalized Online PCA: A Regularized Algorithm with Exact High-Dimensional Dynamics.* arXiv preprint, arXiv:2512.01231, 2025.
- **Samet Demir**, Uras Mutlu, Ozgur Ozdemir. *Neural Academic Paper Generation.* arXiv preprint, arXiv:1912.01982, 2019.

Research Experience

- **Oden Institute, The University of Texas at Austin** Sep 2026 – Present
  Peter O'Donnell, Jr. Postdoctoral Fellow (Advisor: Prof. Blake Bordelon)
  - Research on high-dimensional learning theory (collaboration with Prof. Yue M. Lu)
- **KUIS AI Research Center, Koc University** Sep 2020 – Aug 2026
  Research Assistant (Advisor: Prof. Zafer Dogan)
  - Research on high-dimensional learning theory, with emphasis on nonlinear feature learning
- **Dependable Systems Group, Bogazici University** Mar 2019 – Jul 2020
  Undergraduate Research Assistant (Advisor: Prof. Alper Sen)
  - Research on the robustness of deep learning and adversarial methods
- **PILAB–SIGMA Research Group, Bogazici University** Sep 2018 – Jan 2019
  Undergraduate Research Assistant (Advisor: Prof. A. Taylan Cemgil)
  - Research on Turkish word embeddings with online matrix factorization

Teaching Experience

- **Introduction to Modern Learning Theory, Koc University** Sep 2025 – Jun 2026
  Head Teaching Assistant (Instructor: Prof. Zafer Dogan)
- **Programming for Engineers, Koc University** Sep 2024 – Jun 2026
  Head Teaching Assistant (Instructor: Prof. Zafer Dogan)
- **Algorithms & Complexity, Koc University** Sep 2023 – Jun 2024
  Teaching Assistant (Instructor: Prof. Deniz Yuret)
- **Software Engineering, Koc University** Feb 2023 – Jun 2023
  Teaching Assistant (Instructor: Prof. Attila Gursoy)
- **Programming with Python, Koc University** Sep 2022 – Jan 2023
  Teaching Assistant (Instructor: Prof. Cigdem Gunduz Demir)
- **Computer Systems and Programming, Koc University** Sep 2020 – Jun 2022
  Teaching Assistant (Instructor: Prof. Aykut Erdem)
- **Introduction to Computing, Bogazici University** Sep 2017 – Jan 2018
  Undergraduate Teaching Assistant

Supervision Experience

- **Supervising M.S. Students — four papers so far** Spring 2025 – Present
  Students: Oguzhan Gultekin, Onat Ure, Yigit Yildirim, Ilke Genc

Industry Experience

- **Machine Learning and Artificial Intelligence Intern** Jan 2019 – Mar 2019
  armut.com
- **Software Engineering and Business Analysis Intern** Aug 2019 – Sep 2019
  Intertech
- **Software Engineering Intern** Aug 2018 – Sep 2018
  TUBITAK (The Scientific and Technological Research Council of Turkey)

Honors & Awards

- **Peter O'Donnell, Jr. Postdoctoral Fellowship** 2026–27
  Oden Institute, UT Austin
- **AI Fellowship** Fall 2020 – 2026
  KUIS AI Research Center
- **Ph.D. Scholarship** Fall 2020 – 2026
  TUBITAK (The Scientific and Technological Research Council of Turkey)
- **ICML 2026 Financial Aid Award** 2026
  International Conference on Machine Learning

- **NeurIPS 2025 Scholar Award** 2025
  Neural Information Processing Systems Foundation
- **ICLR 2025 Financial Assistance Award** 2025
  International Conference on Learning Representations
- **Travel Grant** 2023, 2024, 2025, 2026
  Graduate School of Sciences and Engineering, Koc University
- **Teaching Assistant Award** Fall 2024
  Graduate School of Sciences and Engineering, Koc University
- **Honor Certificate** 2020
  B.S., Computer Engineering, Bogazici University

Academic Service (Peer-Reviewing)

- **Neural Information Processing Systems (NeurIPS)** 2025, 2026
- **International Conference on Machine Learning (ICML)** 2026
- **International Conference on Learning Representations (ICLR)** 2025, 2026, 2027
- **Transactions on Machine Learning Research (TMLR)** 2026–
- **AAAI Conference on Artificial Intelligence (AAAI)** 2026
- **IEEE Int. Conf. on Acoustics, Speech and Signal Processing (ICASSP)** 2026
- **Asian Conference on Machine Learning (ACML)** 2024, 2025

References

- **Yue M. Lu** — *Prof., Harvard University*
- **Blake Bordelon (Postdoctoral Advisor)** — *Asst. Prof., Oden Institute, UT Austin*
- **Zafer Dogan (Ph.D. Advisor)** — *Asst. Prof., Koc University*